

SE 216 – SOFTWARE PROJECT MANAGEMENT
SOFTWARE TOOLS DOCUMENT

PROJECT NAME: AI Personal Health & Fitness Coach

GROUP MEMBERS: Mehmet Kahraman, Kerem Güneş, Utku Sertkaya, İsmail Yiğit Şendur, Onur Açar, Salikh İbragimov

TASK #	PROJECT TASKS WHICH REQUIRE SOFTWARE TOOL SUPPORT
1	Development of the AI Personalized Recommendation & Optimization Engine
2	UI/UX Design for Interactive Workout Dashboards & Social Leaderboards
3	Scalable Backend Infrastructure & Health Data Security Management
4	Development Environment Setup for AI and Backend Implementation

SOFTWARE TOOLS FOR TASK 1:

Tool Cost/Training/Functionality Data

Tool	Hugging Face	vLLM	AWS SageMaker	Azure AI Studio
Cost	\$0 (Open Source)	\$0 (Open Source)	\$500/mo	\$420/mo
Training Days	5	3	12	10
Functionality	10	6	8	9

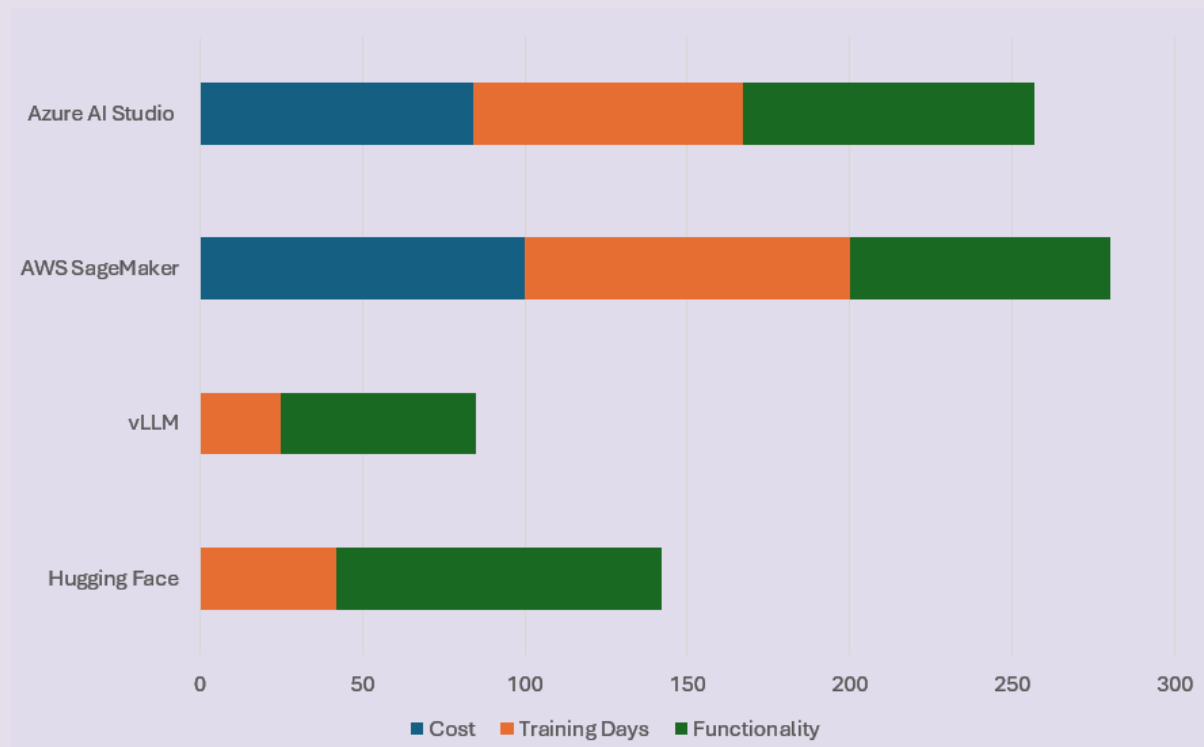
SE 216 – SOFTWARE PROJECT MANAGEMENT

SOFTWARE TOOLS DOCUMENT

Normalized Cost/Training/Functionality Data

Tool	Hugging Face	vLLM	AWS SageMaker	Azure AI Studio
Cost	0	0	100	84
Training Days	42	25	100	83
Functionality	100	60	80	90

Normalized Tool Graph



Which tool has been selected? Why?

Hugging Face was selected.

Why: Getting pre-trained models up and running quickly was our top priority here, alongside painless fine-tuning. Hugging Face ended up being the obvious pick. Rather than spending almost two weeks wrestling with massive cloud environments like SageMaker just to deploy a tweaked model, their ecosystem lets us pull pre-trained weights, fine-tune them on our custom dataset, and get everything live in a fraction of the time. Plus, keeping the

SE 216 – SOFTWARE PROJECT MANAGEMENT
SOFTWARE TOOLS DOCUMENT

base cost at zero while still getting top-tier usability gives the team way more breathing room when we inevitably need to scale up the infrastructure later.

SOFTWARE TOOLS FOR TASK 2:

Tool Cost/Training/Functionality Data

Tool	Figma	Adobe XD	Sketch	InVision
Cost	\$15/mo	\$23/mo	\$12/mo	\$8/mo
Training Days	3	5	4	2
Functionality	10	8	7	6

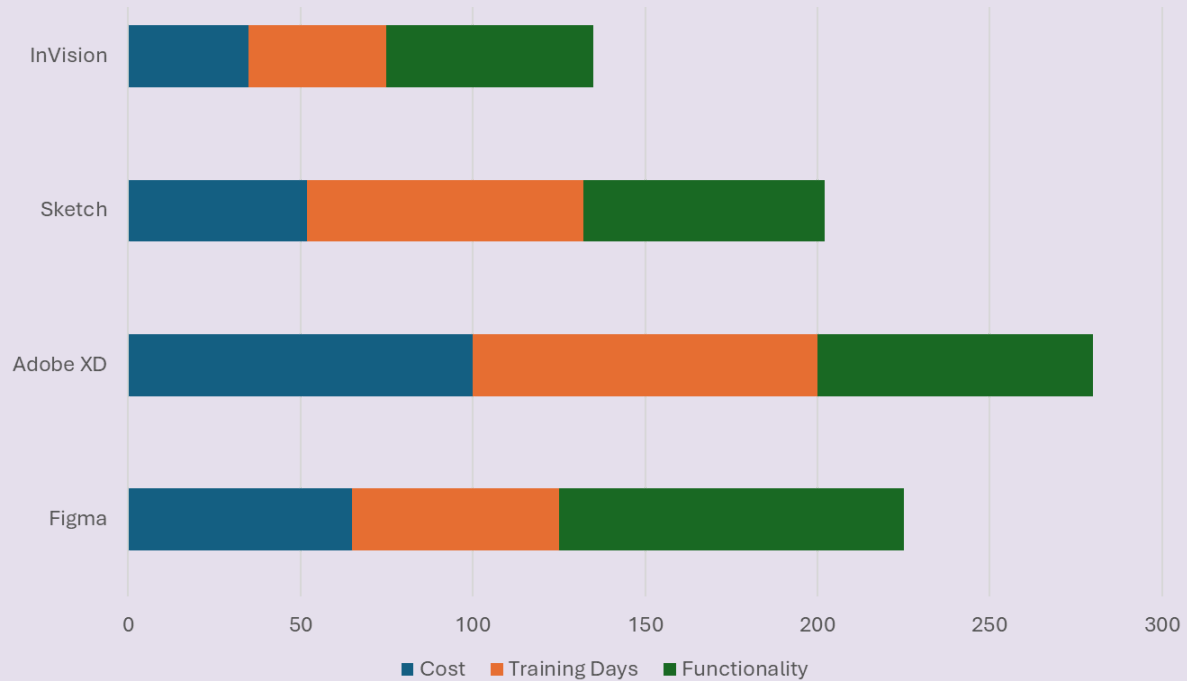
Normalized Cost/Training/Functionality Data

Tool	Figma	Adobe XD	Sketch	InVision
Cost	65	100	52	35
Training Days	60	100	80	40
Functionality	100	80	70	60

SE 216 – SOFTWARE PROJECT MANAGEMENT

SOFTWARE TOOLS DOCUMENT

Normalized Tool Graph



Which tool has been selected? Why?

Figma was selected.

Why: Since the system requires high-fidelity exercise illustrations and a social leaderboard with profile customization, Figma's superior collaborative features and vast plugin library for mobile UI components make it the most efficient choice for a high-quality user experience.

SE 216 – SOFTWARE PROJECT MANAGEMENT
SOFTWARE TOOLS DOCUMENT

SOFTWARE TOOLS FOR TASK 3:

Tool Cost/Training/Functionality Data

Tool	Firebase	PostgreSQL	MongoDB Atlas	Microsoft Azure SQL
Cost	25/mo	0 (Open Source)	30/mo	120/mo
Training Days	3	10	5	12
Functionality	8	9	8	10

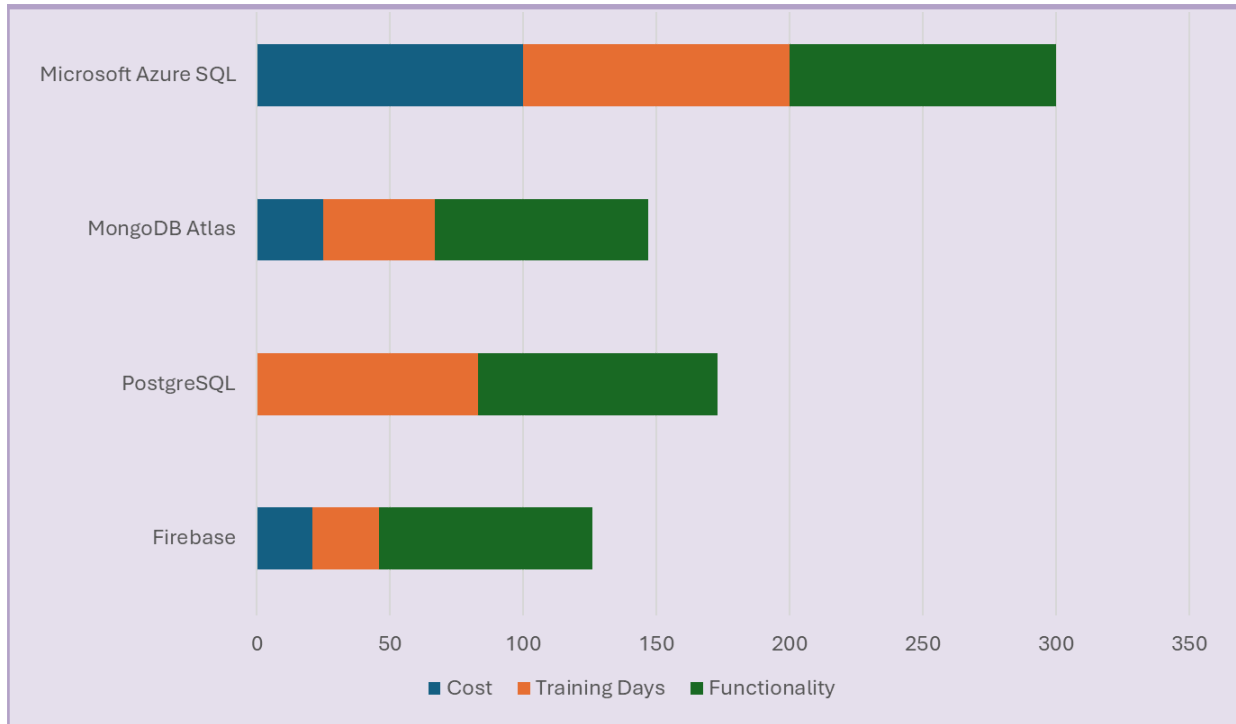
Normalized Cost/Training/Functionality Data

Tool	Firebase	PostgreSQL	MongoDB Atlas	Microsoft Azure SQL
Cost	21	0	25	100
Training Days	25	83	42	100
Functionality	80	90	80	100

Normalized Tool Graph

SE 216 – SOFTWARE PROJECT MANAGEMENT

SOFTWARE TOOLS DOCUMENT



Which tool has been selected? Why?

Firestore (Google Cloud) was selected.

Why: For a health app requiring real-time updates (leaderboards) and secure user authentication (medical data/warnings), Firestore provides an all-in-one "Backend-as-a-Service" solution. It minimizes training time and infrastructure setup, allowing the development team to focus on the AI features rather than server maintenance.

SOFTWARE TOOLS FOR TASK 4:

Tool Cost/Training/Functionality Data

Tool	PyCharm Professional	Wing IDE Pro	Eclipse	Visual Studio
Cost	\$1500/yearly	\$690	\$0	\$3240/yearly
Training Days	6	5	8	4
Functionality	10	8	7	8

SE 216 – SOFTWARE PROJECT MANAGEMENT
SOFTWARE TOOLS DOCUMENT

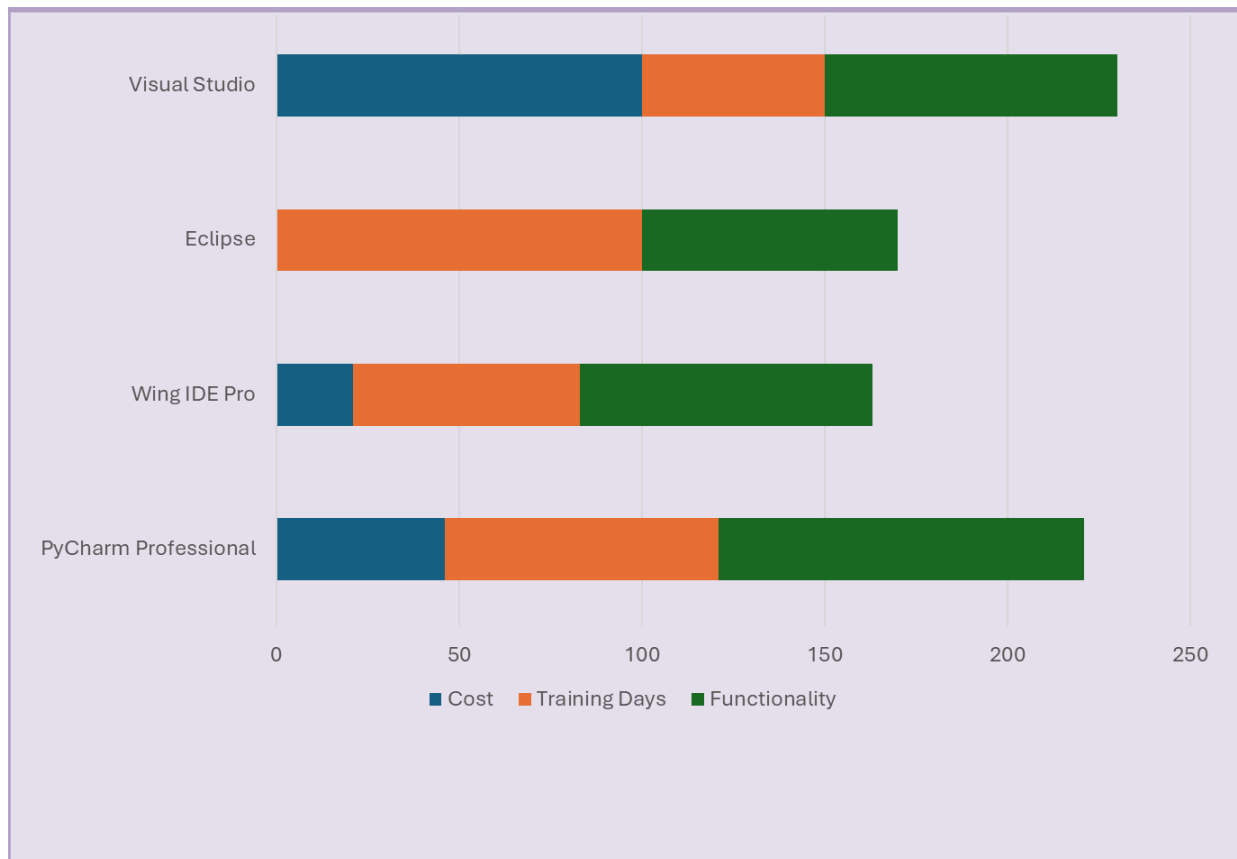
Normalized Cost/Training/Functionality Data

Tool	PyCharm Professional	Wing IDE Pro	Eclipse	Visual Studio
Cost	46	21	0	100
Training Days	75	62	100	50
Functionality	100	80	70	80

Normalized Tool Graph

SE 216 – SOFTWARE PROJECT MANAGEMENT

SOFTWARE TOOLS DOCUMENT



Which tool has been selected? Why?

PyCharm Professional was selected as the development environment for Python because it provides advanced features such as intelligent code completion, powerful debugging tools, integrated testing support, database management, and seamless integration with AI and backend frameworks. Although it is a paid solution, its high functionality and enterprise-level capabilities make it the most suitable IDE for large-scale and AI-based projects.